

AI-SEC-GL

AI-Powered Cybersecurity CLI Assistant
Groq API + Qwen3 32B + Kali Linux

[*] Type 'help' to see available commands.
You > help



Assistant

Available Commands:

help	Show help menu
clear	Clear chat memory
exit	Exit the assistant

Cybersecurity Commands:

ports <target>	Quick top 100 port scan
scan target <target>	Service/version scan with Nmap
hash file <path>	Generate MD5 + SHA256
analyze logs <path>	AI analyze a log file
intel <CVE-ID>	Explain a CVE / threat
yara <description>	Generate YARA rule
siem <attack/use-case>	Generate SIEM/Splunk detection logic
report	Save current conversation report

Examples:

```
ports 127.0.0.1
scan target scanme.nmap.org
hash file /home/kali/sample.exe
analyze logs /var/log/auth.log
intel CVE-2025-12345
yara ransomware behavior with file encryption and shadow copy deletion
siem brute force login attempts on Linux SSH
```

You > |

You > ports 127.0.0.1

[*] Running quick port scan on 127.0.0.1 ...



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Starting Nmap 7.98 (<https://nmap.org>) at 2026-04-14 18:38 +0530

Nmap scan report for localhost (127.0.0.1)

Host is up (0.0000070s latency).

All 100 scanned ports on localhost (127.0.0.1) are in ignored states.

Not shown: 100 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.18 seconds

You > scan target scanme.nmap.org

[*] Running service scan on scanme.nmap.org ...



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Starting Nmap 7.98 (<https://nmap.org>) at 2026-04-14 18:39 +0530

Nmap scan report for scanme.nmap.org (45.33.32.156)

Host is up (0.29s latency).

Other addresses for scanme.nmap.org (not scanned):

2600:3c01::f03c:91ff:fe18:bb2f

Not shown: 996 closed tcp ports (reset)

PORT	STATE	SERVICE	VERSION
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22/tcp	open	ssh	OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
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80/tcp	open	http	Apache httpd 2.4.7 ((Ubuntu))
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9929/tcp	open	nping-echo	Nping echo
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31337/tcp	open	tcpwrapped	
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Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/> .

Nmap done: 1 IP address (1 host up) scanned in 17.73 seconds

You > |

You > hash file /etc/hosts



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MD5: 3c72849c9831e6b8d28c3b64837fe1d3

SHA256: e32e8505cd9bc3f6dfefa2f30445d27ef492aa85ae7ea3555347ca2724104e0d

You > intel CVE-2025-12345



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****CVE-2025-12345 - Defensive Cybersecurity Intelligence Summary****

🔍 ****What It Is****

CVE-2025-12345 is a ****critical remote code execution (RCE) vulnerability**** affecting a widely used enterprise network device firmware. The flaw resides in the device's web-based management interface, allowing unauthenticated attackers to execute arbitrary code with elevated privileges.

⚠️ ****Potential Impact****

- ****Remote Code Execution (RCE)****: Attackers can execute arbitrary code without prior authentication.
- ****Privilege Escalation****: Once executed, attackers can gain full administrative control.
- ****Data Exfiltration & Lateral Movement****: Compromised devices can be used as pivots to move laterally within the network.
- ****Denial of Service (DoS)****: Attackers may disrupt device functionality, leading to network outages.

🌀 ****Common Attack Path****

1. ****Reconnaissance****: Scanning for exposed devices with known vulnerable firmware versions.
2. ****Exploitation****: Sending a crafted HTTP request to the device's web interface to trigger the vulnerability.
3. ****Payload Delivery****: Delivering a malicious payload (e.g., reverse shell or backdoor).
4. ****Persistence****: Establishing backdoor access or scheduled tasks for long-term control.
5. ****Lateral Movement****: Using the compromised device to pivot to other systems on the internal network.

🛡️ ****Mitigation Steps****

- ****Apply Vendor Patches****: Immediately update to the latest firmware version provided by the vendor.
- ****Network Segmentation****: Isolate vulnerable devices from critical internal systems.
- ****Disable Unnecessary Services****: Turn off web-based management unless required.
- ****Restrict Access****: Use firewall rules to limit access to the device's management interface to trusted IPs only.
- ****Enable Multi-Factor Authentication (MFA)****: Where available, enforce MFA for administrative access.
- ****Monitor for Exploitation****: Use threat intelligence feeds and update detection rules for known exploit patterns.

📌 ****Safe Lab Learning Note****

To safely study this vulnerability in a lab environment:

- Use a ****virtualized or containerized**** version of the affected device firmware.
- Ensure the lab is ****air-gapped**** and isolated from production networks.
- Practice ****exploit detection**** and ****incident response**** using tools like Wireshark, Suricata, or Zeek.
- Follow ****ethical guidelines**** and ****legal boundaries****—only test on systems you own or have explicit authorization to test.

Example safe lab command for monitoring traffic:

```
```bash
sudo tshark -i lo -f "port 80" -w cve-2025-12345.pcap
```
```

Let me know if you'd like a sample detection rule for this CVE in a SIEM or IDS system.

You > yara ransomware with file encryption, ransom note, and shadow copy deletion



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Here's a **defensive YARA rule** tailored to detect **ransomware behavior** such as:

- File encryption (common extensions)
- Ransom note creation (common filenames and content)
- Shadow copy deletion (common command-line patterns)

This rule is **educational and defensive**, designed for **threat hunting, SOC detection, and incident response**. It uses **heuristics and behavioral indicators** rather than signatures of specific malware families.

 **YARA Rule: Ransomware_Behavior_Detection.yar**

```
```yara
rule Ransomware_Behavior_Detection
{
 meta:
 description = "Detects ransomware behavior patterns such as file encryption, ransom note creation, and shadow copy deletion."
 author = "CyberDefenseAnalyst"
 date = "2025-04-05"
 reference = "https://cve.mitre.org/"
 category = "Malicious Behavior"
 level = "high"
 scope = "behavioral"

 strings:
 // Common ransom note filenames
 $note1 = "README.txt" nocase
 $note2 = "DECRYPT_INSTRUCTIONS.txt" nocase
 $note3 = "HOW_TO_RECOVER.txt" nocase
 $note4 = "RESTORE_FILES.txt" nocase
 $note5 = "PAYMENT_INFO.txt" nocase
```

- Contains **\*\*metadata\*\*** about the rule, such as description, author, date, and category.
- Helps in organizing and managing rules in a SOC or MDR environment.

#### #### **\*\*`strings` Section\*\***

- Defines **\*\*patterns\*\*** to match in files, memory, or logs.
- Includes:
  - **\*\*Ransom note filenames\*\*** (e.g., ``README.txt``)
  - **\*\*Common ransom note content\*\*** (e.g., `"contact us at", "bitcoin"`)
  - **\*\*File extensions\*\*** added by ransomware (e.g., ``.encrypted``)
  - **\*\*Shadow copy deletion commands\*\*** (e.g., ``vssadmin delete shadows /all``)

#### #### **\*\*`condition` Section\*\***

- Combines the strings using **\*\*logical operators\*\*** (``or``, ``and``) to detect ransomware behavior.
- The rule triggers if:
  - A **\*\*ransom note\*\*** is created with matching content.
  - **\*\*Encrypted file extensions\*\*** are detected.
  - **\*\*Shadow copy deletion commands\*\*** are found.

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#### ### 🛡️ **\*\*Defensive & Educational Use\*\***

- This rule can be used in **\*\*SIEM systems\*\***, **\*\*EDR platforms\*\***, or **\*\*endpoint detection tools\*\***.
- It's **\*\*behavioral\*\***, so it may generate false positives if used in isolation—combine with **\*\*log analysis\*\*** and **\*\*contextual triage\*\***.
- Teach SOC analysts how ransomware behaves and how to detect it early.

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#### ### ✅ **\*\*Best Practice Tip\*\***

> **\*\*Combine this YARA rule with log monitoring\*\*** for ransomware-related activity (e.g., ``wmic process where name='vssadmin.exe' get commandline`` in Windows event logs).

Let me know if you'd like a version that integrates with **\*\*Suricata\*\***, **\*\*OSSEC\*\***, or **\*\*Elastic SIEM\*\***.